

**STAT 404 Project Report:**

**Exploring the Effects of Soaking Boiled Eggs in Various Liquids on Weight Changes**

*A Study of Osmosis and Chemical Interactions in Food Science*

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## 1 Abstract

In today's world, food costs are steadily rising, thus preserving food to extend its shelf life has become increasingly important. Understanding the processes that affect food quality and longevity is a key focus in food science. This experiment investigates how soaking boiled eggs in solutions with different pHs (acidic and basic) under varying conditions impacts their weight change. The focus of this study is to explore the effects of liquid type, eggshell condition, refrigeration, and soak duration on the eggs, particularly through osmosis and chemical interactions. By examining these factors, we aim to gain insights into food preservation methods, which can help optimize storage practices and improve the longevity of food items.

## 2 Introduction

We chose medium white eggs under two soaking durations (1 hour and 3 hours) to run the experiment on. The tests will be run with varying treatment factors such as liquid type, eggshell condition, and refrigeration, on weight change, particularly through osmosis and chemical interactions. We also wish to examine the impact on egg composition, particularly how soaking time and eggshell condition affect the permeability of the egg to the liquids. Moisture content and water activity are crucial in understanding microbial growth (Gichau et. al, 2019). Hence, we gauge the moisture absorption through the percentage of weight change - our response - that occurs between the experimental unit pre and post application of treatment. Our experiment follows a  $2^3$  fractional factorial design spread across two blocks. We have 3 treatment factors and 1 blocking factor with a total of 8 runs.

The following sections will cover the details of our experimental design in Section 3, followed by a statistical analysis in Section 4, and the conclusion of our experiment in Section 5. Then in the appendix in Section 6, we have Section 6.1 for tables and figures and Section 6.2 for details of our data.

## 3 Details of the Experimental Design

We used eight medium white eggs to complete this experiment with varying treatment factors applied and recorded the weight change in percentage. Below is our model and the factors and their respective levels are summarized in Table 3.1. We also have Table 6.2.1. in the Appendix that better describes how we applied each treatment and block to our runs.

$$Y_{ijkl} = \mu + \alpha_j + \beta_k + \gamma_l + \delta_i + (\alpha\beta)_{jk} + (\alpha\gamma)_{jl} + \epsilon_{ijkl}$$

Figure 3.1 Model with 3 main effects, 2 two-level interaction effects, and one blocking effect (aliased with three-level interaction)

Figure 3.1 notes the model we use throughout this study.  $\mu$  here represents the overall effect of the model.  $\alpha_j$  denotes the main effect for Liquid Type (LT) for level  $j$ , where  $j$  is  $\{1, 2\}$ . Similarly,  $\beta_k$  is the coefficient for the main effect of Eggshell (E) for level  $k$ , where  $k$  is  $\{1, 2\}$ ,  $\gamma_l$  is the main effect for Refrigeration (R) for level  $l$ , where  $l$  is  $\{1, 2\}$  and  $\delta_i$  is the main effect for Soaking Duration (block) for level  $i$ , where  $i$  is  $\{1, 2\}$ . The two factor interactions for  $(\alpha\beta)_{jk}$  and  $(\alpha\gamma)_{jl}$  represent LT:E and LT:R interactions for level  $j$  and  $k$  and level  $j$  and  $l$  respectively.  $\epsilon_{ijkl}$  is the i.i.d error term for level  $i, j, k, l \sim N(0, \sigma^2)$

Table 3.1 below provides a comprehensive look at our factors and their levels.

| Variables          | Description                            | Levels                                    |
|--------------------|----------------------------------------|-------------------------------------------|
| LT (Liquid Type)   | Type of liquid the egg is soaked in    | Basic = +1,<br>Acidic = -1                |
| E (Eggshell)       | Whether the eggshell is peeled or not  | On (Unpeeled) = +1,<br>Off (Peeled) = -1  |
| R (Refrigerated)   | Whether the soaked egg is refrigerated | Refrigerated = +1,<br>Unrefrigerated = -1 |
| SD (Soak Duration) | Time the egg is soaked in the liquid   | 1 hour = -1,<br>3 hours = +1              |

Table 3.1: Factors and Levels

The response variable is the percentage change in egg weight, calculated as:

$$\text{Percentage Change in Weight (\%)} = \frac{(\text{Final Weight} - \text{Initial Weight})}{\text{Initial Weight}} \times 100$$

Measurement Device: Digital kitchen scale (accuracy:  $\pm 0.1$  g).

Note: The order in which the eggs are soaked in each solution, the assignment of the egg conditions, and whether the egg will be refrigerated or not, will be randomized. Randomization eliminates bias and ensures that the results are not influenced by any uncontrolled factors. The Weight Difference (%) was also put through an absolute function to maintain uniformity of weight change (we are concerned with the magnitude of difference, rather than the positivity or negativity of the percentage).

| Runs | LT | E | R | LT : E | LT : R | E : R | SD = LT : E : R | Weight Difference (%) |
|------|----|---|---|--------|--------|-------|-----------------|-----------------------|
| 1    | -  | - | - | +      | +      | +     | -               | 2.87                  |
| 2    | -  | - | + | +      | -      | -     | +               | 4.30                  |
| 3    | -  | + | - | -      | +      | -     | +               | 1.70                  |
| 4    | -  | + | + | -      | -      | +     | -               | 0.72                  |
| 5    | +  | - | - | -      | -      | +     | +               | 0.88                  |
| 6    | +  | - | + | -      | +      | -     | -               | 0.89                  |
| 7    | +  | + | - | +      | -      | -     | -               | 0.19                  |
| 8    | +  | + | + | +      | +      | +     | +               | 1.79                  |

Table 3.2. Contrast matrix for a  $2^3$  experiment in two blocks of four runs

## 4 Statistical Analysis

The statistical analysis will use analysis of variance (ANOVA) to assess the main effects of Liquid Type, Eggshell Condition, Refrigeration, and Soak Duration, as well as some two-factor interactions. Once the model is set up, various hypotheses for coefficients of each factor will be tested to comprehend their significance. Model assumptions will be evaluated using visualizations.

### 4.1 Model Setup

We use a half fractional factorial design with 3 treatment factors and 1 blocking factor, with a total of 8 runs. In our design generator in Table 6.1.7., we are shown that the main effects are aliased with the three-factor interaction effects, and the two-factor interaction effects are confounded with other two-factor interaction effects. Moreover, the defining relation's shortest word is  $I = 123B$ , thus our design is Resolution IV.

According to Figure 6.1.7, which shows the mean weight difference between the two levels for each of our factors, Liquid Type (LT) varies the most between the two levels. Hence, we include the two-factor interactions between Liquid Type (LT) and the other two treatment factors (Egg Condition and Refrigeration). Our inclusion of these terms in the model is further corroborated by the interaction plot in Figure 6.1.3 which demonstrates the relationship between Liquid Type (LT) and whether the eggs are Refrigerated (R) and Figure 6.1.4 showcasing the relationship between Liquid Type (LT) and Egg Condition (E). In Figure 6.1.3, the lines do not appear parallel, indicating an interaction between these two factors. Specifically, Refrigerated eggs show a positive weight difference, while Non-Refrigerated eggs show a negative one. This suggests that Liquid Type has a greater impact on refrigerated eggs than on non-refrigerated ones. In Figure 6.1.4, we can see that the effect of Liquid Type on weight change is dependent on whether the eggshell is peeled. For unpeeled eggs, the weight difference is more significant between the two liquid types, while for peeled eggs, the effect is less pronounced, indicating that the eggshell condition influences how the liquid types affect the weight change. Only these two-factor interactions were chosen to retain some degrees of freedom for the residual.

Soaking Duration is used as a blocking factor due to its already proven effect on percentage of weight change. For eggs that are soaked for longer periods, we expected to see greater magnitude of change as more time allowed for more chemical osmotic activity. Due to this blocking, we were able to investigate and focus on the estimated effects of our treatment factors in more depth.

### 4.2 Model Checking and Improvement

Initially, we complete a Box-Cox analysis before proceeding with the ANOVA statistical inference. This is shown in Figure 6.1.2. and we can see that  $\lambda = 1$  is not in the 95% confidence interval, which suggests our model requires a transformation. We applied a square root transformation to reduce the right skewness.

The assumptions of the ANOVA model will be checked using diagnostic plots to ensure that the model fits appropriately. If any assumptions are violated, transformations or alternative statistical methods may be considered. Specifically, to check the normality of the residuals, we plot a Residuals vs Fitted Values plot as shown in Figure 6.1.8. Since no pattern is detected, we can be confident that the errors are random with equal variance. Figure 6.1.9 shows the QQ plot where we notice that the values fall flat horizontally. This suggests that due to the scale of the response and low variability within our runs, our errors don't have sufficient data variability to form a distribution let alone a Normal Distribution. This is worrisome but cannot be avoided due to the limitations of our resources. Hence, we assume normality of our errors following a  $N(0, \sigma^2)$  distribution and their independence (i.i.d) moving forward.

### 4.3 ANOVA

To test the hypothesis regarding the significance of all our terms, we performed ANOVA on our model. Our null hypothesis ( $H_0$ ) stated that the said estimated coefficient was 0, whereas our alternative ( $H_a$ ) argued that it was not equal to 0.

Figure 6.1.1 tabulates the results of ANOVA for our model. Based on those results, all of our estimated coefficients are statistically significant at 5% level due to their p-values being very low, which leads us to reject all null hypotheses to conclude that there is not sufficient evidence for the beta terms to be null.

Out of the coefficients, Liquid Type (LT) seems to be the most significant term with the lowest p-value, indicating that pH of the solution is highly influential on the percentage of weight change. Additionally, the interaction term LT:E is the most significant two-factor interaction. Since the blocking term was statistically significant at 5%, we can also conclude that blocking in this treatment was useful.

To graphically estimate these coefficients, we visualized the half-normal plot with ANOVA estimates in Figure 6.1.10 to outline the most significant terms. LT and E were highly significant.

Table 6.1.10 demonstrates the mean weight change (%) for each combination of LT and E, which will be useful in analyzing contrasts in the next section.

### 4.4 Contrasts

Contrast for the Main Effect of Liquid Type (LT):

$$\hat{\kappa} = \frac{\sum_{LT=1} y_i}{4} - \frac{\sum_{LT=-1} y_i}{4} = 0.9375 - 2.397 = -1.4595$$

Standard error for contrast:

$$se(\hat{\kappa}) = \sqrt{\frac{MS(Residuals)}{4} + \frac{MS(Residuals)}{4}} = \sqrt{\frac{0.00048}{4} + \frac{0.00048}{4}} = \sqrt{\frac{0.00096}{4}} = 0.0155$$

Further calculations suggest a high t-statistic with 1 df, leading to a p-value of 0.0034 which is statistically significant under 5%. This means that there is evidence to suggest that there is a difference in mean weight change depending on the pH of the liquid used for soaking (acidic or basic). Since we tested only one contrast, there is no need for Bonferroni Correction.

### 4.5 Interpretation of Interaction Effects

The estimated effects of LT show a significant impact on the response variable, with changes in LT corresponding to measurable shifts in the outcome. The interaction between LT and Eggshell Condition (E) indicates that LT affects the response when the eggshell condition is off, suggesting that the effectiveness of certain liquid types is dependent on the condition of the eggshell. Similarly, the LT and Refrigeration (R) interaction shows that LT also influences the response when the eggs are refrigerated. A favorable combination for preserving the eggs is when LT is at level 1 (more effective preservation) and the eggs are refrigerated.

## 5 Conclusions and Discussion

All the main effects and interaction effects were found to be significant at a 5% level. This suggests that all the terms have a strong effect on our response variable. As shown in Figure 6.1.1., we can see that Liquid Type is the most statistically significant with a p-value of 0.017, indicating liquid type has a major effect on the response. Further, its contrast in 4.4 denotes that there is a significant difference between each level (acidic or basic) at 5% significance. Contrastingly, Refrigeration has the greatest p-value of 0.047, which suggests it will not have as pronounced an effect on the response.

The observed decrease in weight change suggests that moisture is being drawn out of the eggs, likely due to increased osmosis. This loss of moisture reduces the likelihood of mold growth, as less moisture in the eggs creates a less hospitable environment for microbial activity. As a result, the preservation of the eggs is improved, extending their shelf life. These findings highlight the potential for enhanced food preservation methods that could reduce food waste and help lower food costs, particularly benefiting low-income individuals who are more affected by rising food prices.

Due to the lack of resources, our data collection was limited which led to a fractional factorial design. With an abundance of resources, our experimental design can be improved by the addition of replication to expand the degrees of freedom for our residuals. This way more two-factor interaction effects and three-factor interaction terms could be estimated, preventing aliasing and confounding of terms, and a full factorial experiment can be run.

Another limitation we faced was the scale of our response was very small. This led to observations being similar leading to low variability within the data gathered; this could have led to skewed results and inaccurate conclusions due to low sample size. This was particularly noticed in Figure 6.1.9 where the Normal quantiles fell horizontally due to the low variability within runs/observations, hence making it hard to establish the normality of the errors.

In the presence of more laboratory equipment, precision of our results could be increased and more responses can be studied rather than just weight absorption. One such response could be to study the shelf-life of these experiments over longer periods of time (months, years) to gauge the extent of effectiveness of the techniques posed in this study. Since soaking in acidic/basic solutions is only one of multiple preservation techniques (similar to brining and marinating), our experimental study can be used as a baseline and blueprint to delve into experiments and testing with other methods (packaging and storage etc.).

6 Appendix

6.1 Tables and Figures

Response: sqrt(y)

|           | Df | Sum Sq  | Mean Sq | F value | Pr(>F)  |   |
|-----------|----|---------|---------|---------|---------|---|
| LT        | 1  | 0.64119 | 0.64119 | 1340.68 | 0.01738 | * |
| E         | 1  | 0.37112 | 0.37112 | 775.99  | 0.02284 | * |
| R         | 1  | 0.08644 | 0.08644 | 180.73  | 0.04727 | * |
| B         | 1  | 0.37478 | 0.37478 | 783.65  | 0.02273 | * |
| LT:E      | 1  | 0.28415 | 0.28415 | 594.13  | 0.02610 | * |
| LT:R      | 1  | 0.12081 | 0.12081 | 252.61  | 0.04000 | * |
| Residuals | 1  | 0.00048 | 0.00048 |         |         |   |

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Signif. codes: 0 '\*\*\*' 0.001 '\*\*' 0.01 '\*' 0.05 '.' 0.1 ' ' 1

Table 6.1.1. ANOVA Table

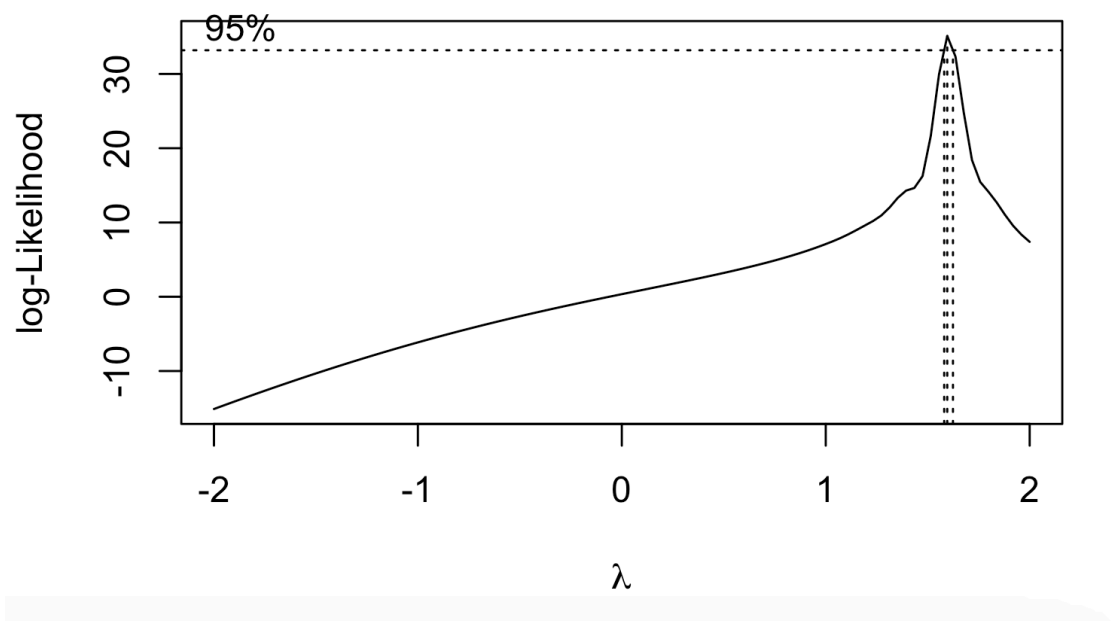


Figure 6.1.2. Box-Cox analysis applied to our model

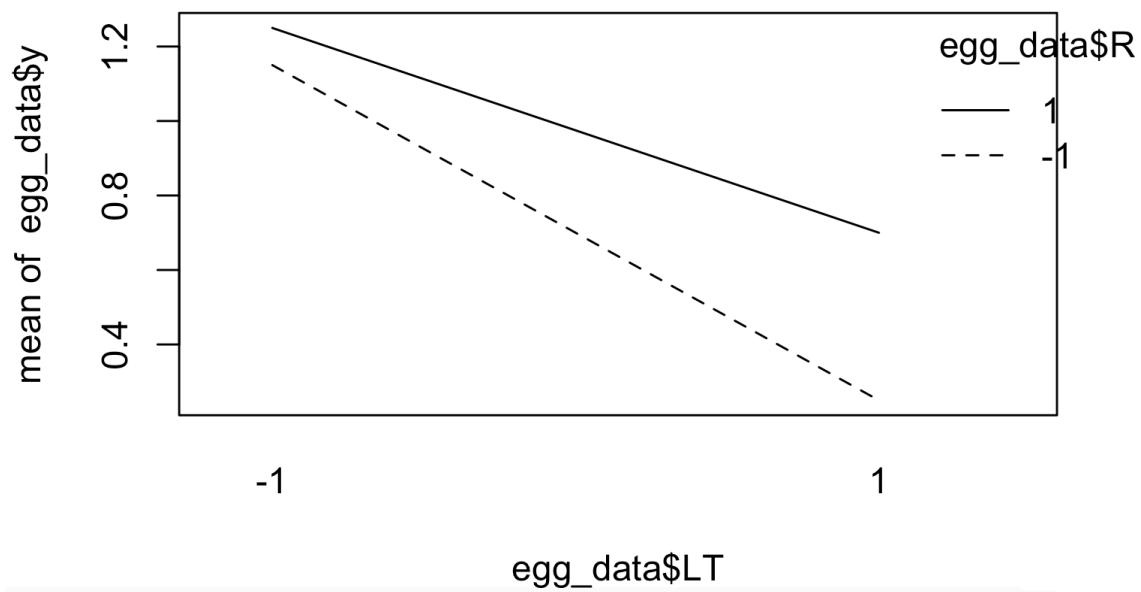


Figure 6.1.3. Mean of weight difference (%) plotted against the two levels of the treatment factor Liquid Type (LT) and the two levels of the treatment factor Refrigerated (R)

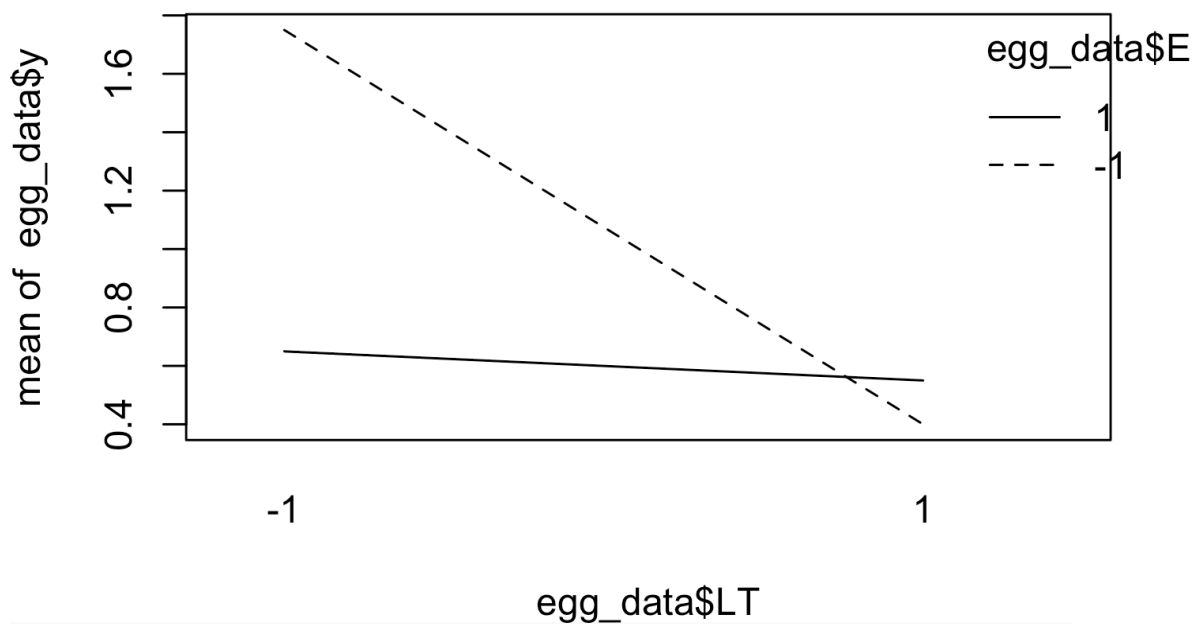


Figure 6.1.4. Mean of weight difference (%) plotted against the two levels of the treatment factor Liquid Type (LT) and the two levels of the treatment factor Eggshell Condition (E)



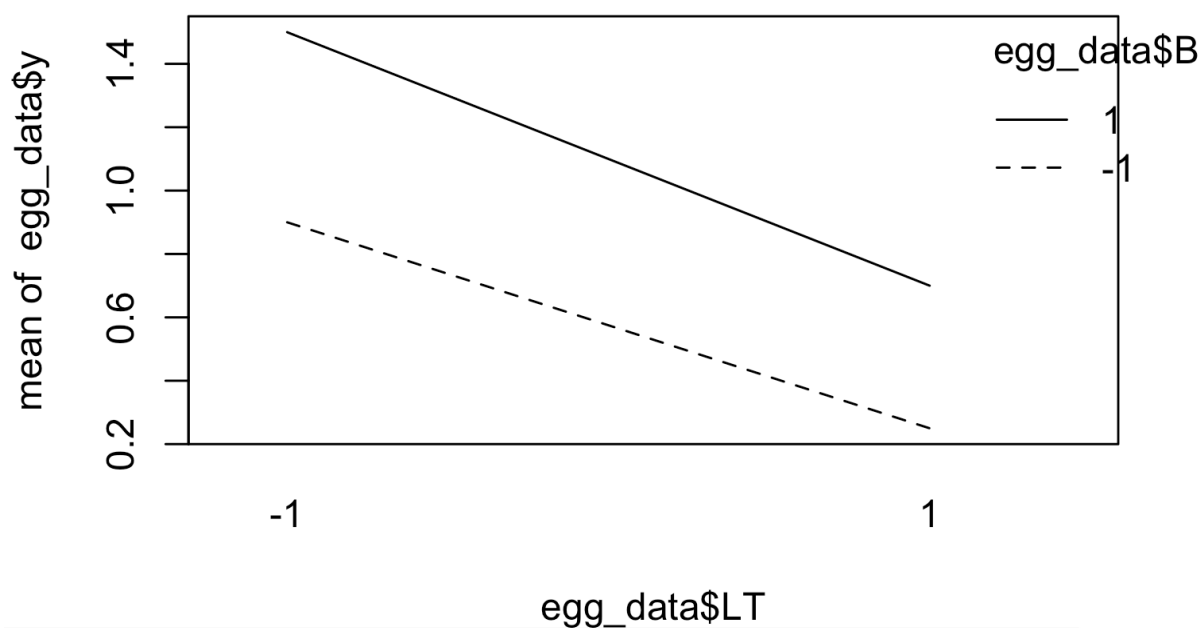


Figure 6.1.5. Mean of weight difference (%) plotted against the two levels of the treatment factor Liquid Type (LT) and the two levels of the blocking factor Soaking Duration (B)

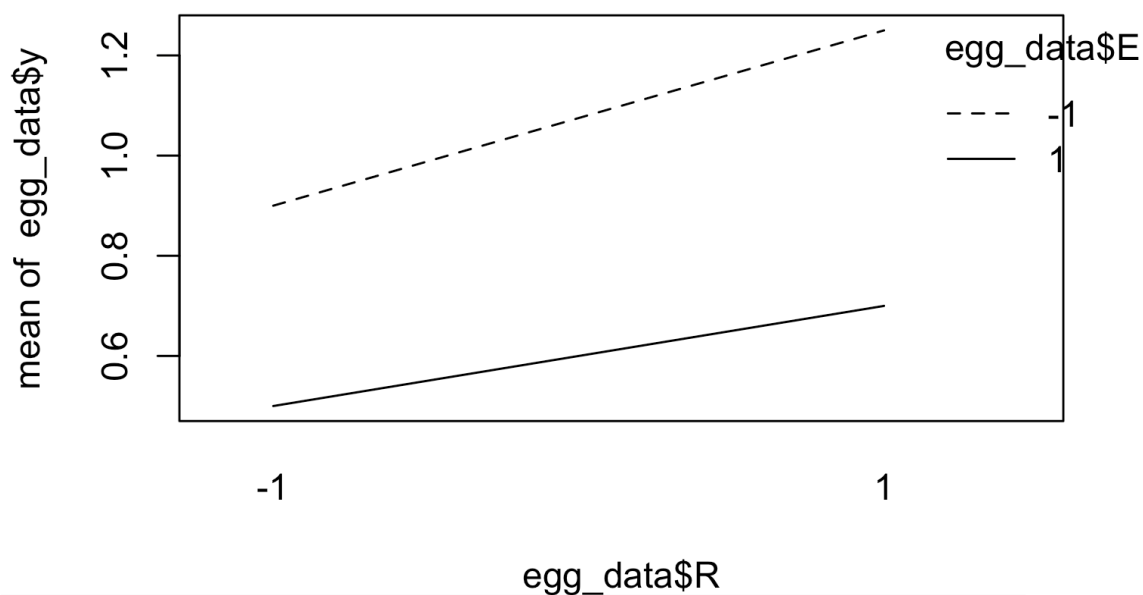


Figure 6.1.6. Mean of weight difference (%) plotted against the two levels of the treatment factor Refrigerated (R) and the two levels of the treatment factor Eggshell Condition (E)

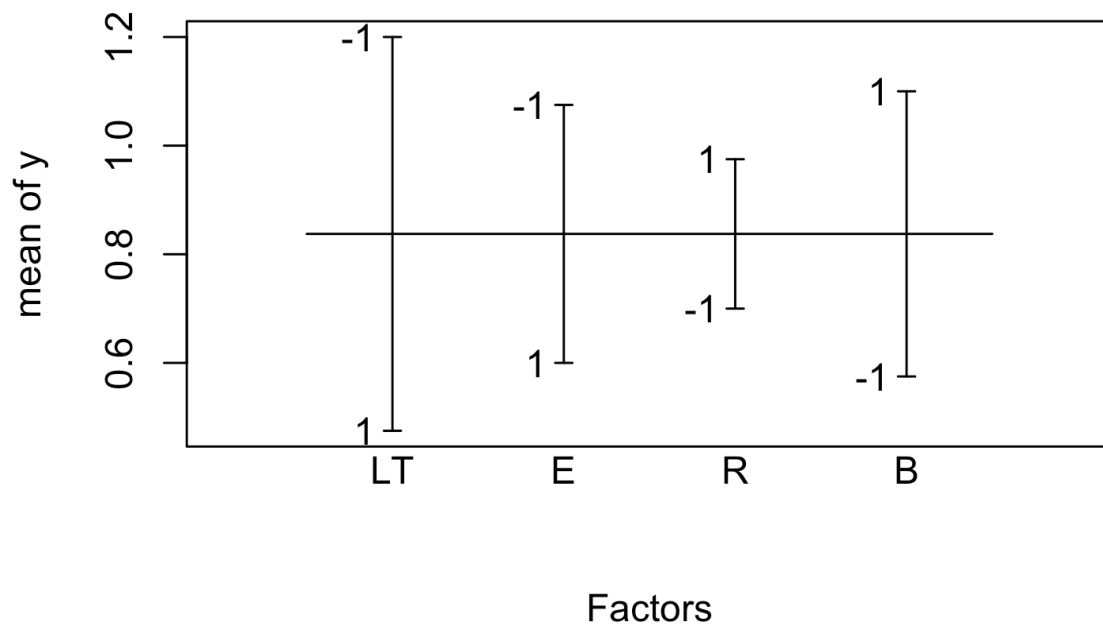


Figure 6.1.7. Design plot: mean of weight difference (%) plotted against the two levels of the treatment factor Liquid Type (LT), the two levels of the treatment factor Eggshell Condition (E), the two levels of the treatment factor Refrigerated (R), and the two levels of the blocking factor Soaking Duration (B).

Design generator B = 123.

1 = LT  
2 = E  
3 = R  
B = SD

Aliases

I = 123B  
1 = 23B  
2 = 13B  
3 = 12B  
12 = 3B  
13 = 2B  
23 = 1B  
123 = B

Table 6.1.7. Design Generators

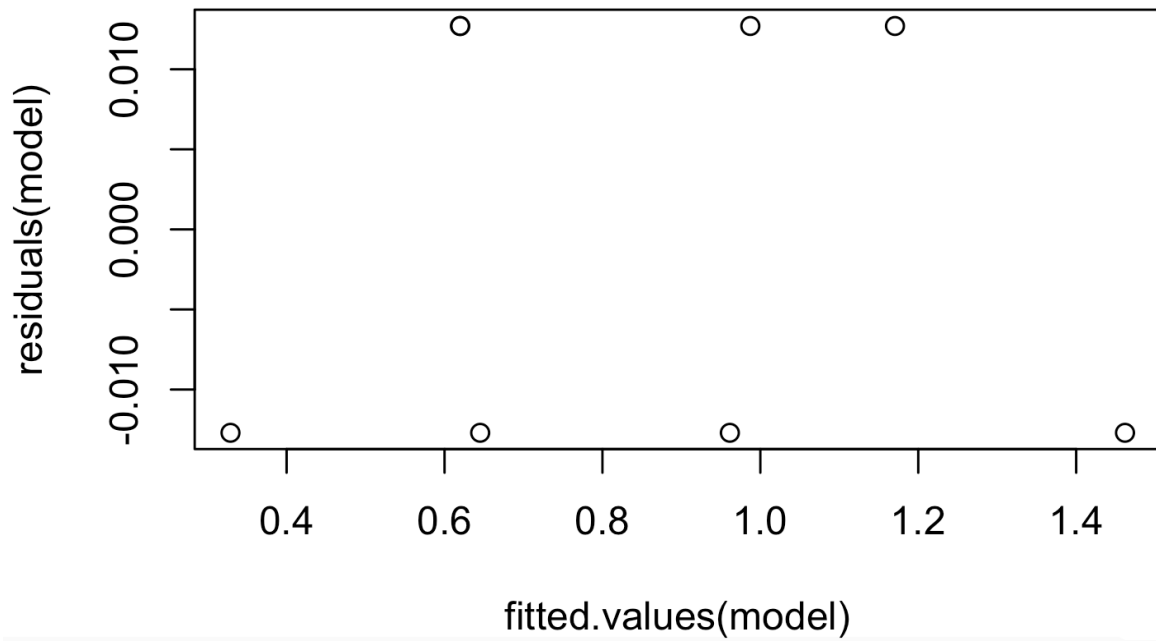


Figure 6.1.8. Residuals vs. Fitted values

### Normal Q-Q Plot

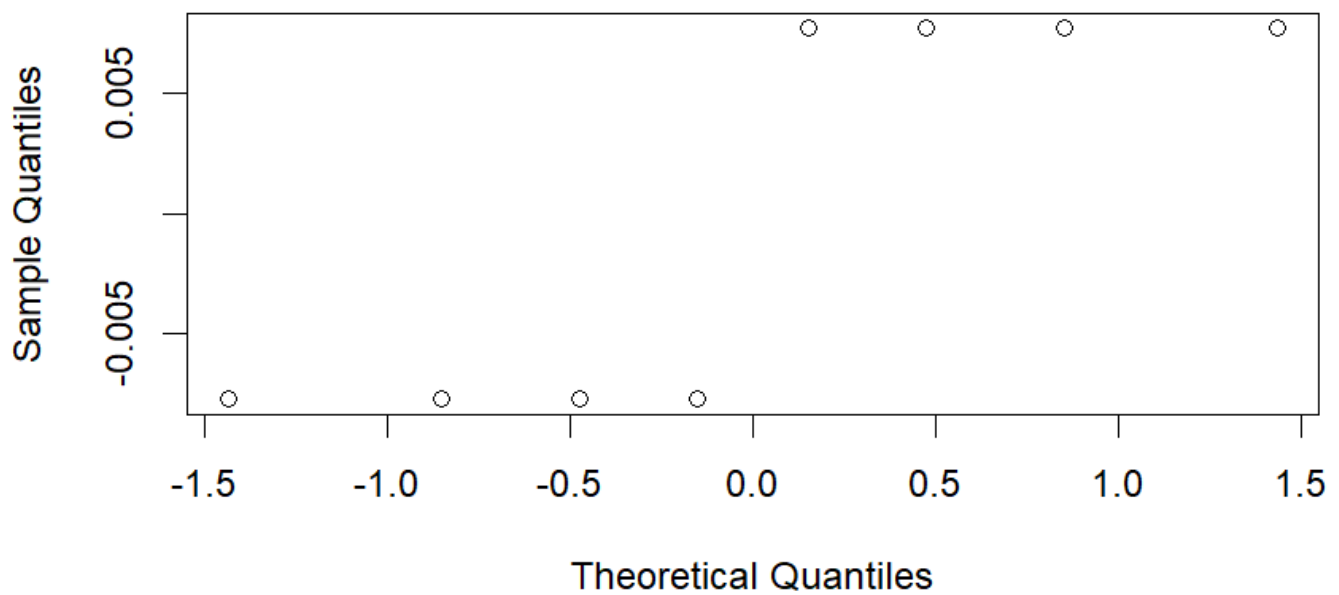


Figure 6.1.9. QQ Plot

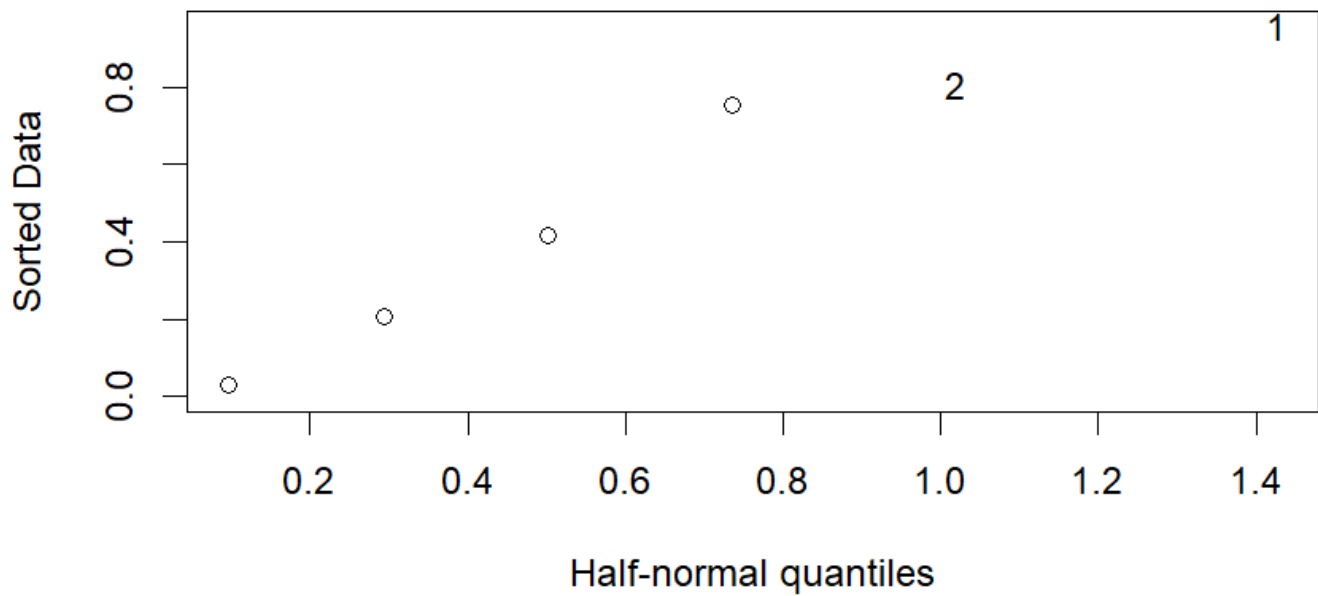


Figure 6.1.10. Half-normal Plot

| LT     | Mean Weight Change (%) |        |
|--------|------------------------|--------|
|        | E (OFF)                | E (ON) |
| Acidic | -3.585                 | 1.210  |
| Basic  | 0.885                  | 0.990  |

Table 6.1.10. Mean Weight Change (%) for combinations of all levels of LT and E

## 6.2 Data Appendix

| Runs | LT     | E   | R                | SD = LT : E : R | Weight Difference (%) |
|------|--------|-----|------------------|-----------------|-----------------------|
| 1    | Acidic | Off | Not Refrigerated | 1 hour          | -2.87                 |
| 2    | Acidic | Off | Refrigerated     | 3 hours         | -4.30                 |
| 3    | Acidic | On  | Not Refrigerated | 3 hours         | 1.70                  |
| 4    | Acidic | On  | Refrigerated     | 1 hour          | 0.72                  |
| 5    | Basic  | Off | Not Refrigerated | 3 hours         | 0.88                  |
| 6    | Basic  | Off | Refrigerated     | 1 hour          | 0.89                  |
| 7    | Basic  | On  | Not Refrigerated | 1 hour          | 0.19                  |
| 8    | Basic  | On  | Refrigerated     | 3 hours         | 1.79                  |

Table 6.2.1. Experimental Runs and the Combination of Factors